

The output shows that we do indeed have a string containing *pi* to 1,000,000 decimal places:

```
3.14159265358979323846264338327950288419716939937510...
1000002
```

Python has no inherent limit to how much data you can work with; you can work with as much data as your system's memory can handle.

NOTE

To run this program (and many of the examples that follow), you'll need to download the resources available at https://ehmatthes.github.io/pcc_3e.

Is Your Birthday Contained in Pi?

I've always been curious to know if my birthday appears anywhere in the digits of *pi*. Let's use the program we just wrote to find out if someone's birthday appears anywhere in the first million digits of *pi*. We can do this by expressing each birthday as a string of digits and seeing if that string appears anywhere in `pi_string`:

```
pi_birthday.py --snip--
for line in lines:
    pi_string += line.strip()

birthday = input("Enter your birthday, in the form mmddyy: ")
if birthday in pi_string:
    print("Your birthday appears in the first million digits of pi!")
else:
    print("Your birthday does not appear in the first million digits of pi.")
```

We first prompt for the user's birthday, and then check if that string is in `pi_string`. Let's try it:

```
Enter your birthdate, in the form mmddyy: 120372
Your birthday appears in the first million digits of pi!
```

My birthday does appear in the digits of *pi*! Once you've read from a file, you can analyze its contents in just about any way you can imagine.

TRY IT YOURSELF

10-1. Learning Python: Open a blank file in your text editor and write a few lines summarizing what you've learned about Python so far. Start each line with the phrase *In Python you can.* . . . Save the file as `learning_python.txt` in the same directory as your exercises from this chapter. Write a program that reads the file and prints what you wrote two times: print the contents once by reading in the entire file, and once by storing the lines in a list and then looping over each line.

(continued)